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SPPU M.Sc. Data Science — Free Study Material

Lab: Big Data

DS-561-MJP

Solved Practical Manual (Hadoop · PySpark)

Semester II · Lab (MJP)

This solved practical manual covers the Big Data lab assignments from the official SPPU M.Sc. Data Science syllabus. Each practical states the aim followed by a complete, runnable solution and the expected output. Use it as a reference — type the code yourself and experiment with the parameters to learn effectively.

Requirements: Python 3.x with numpy, pandas, matplotlib, scikit-learn, scipy installed (pip install numpy pandas matplotlib scikit-learn scipy).

Requirements: Hadoop (single-node) and PySpark (pip install pyspark).

Assignment 1: Data ingestion & storage (HDFS)

```
# Load a CSV into HDFS and read it back
hdfs dfs -mkdir -p /data/ingest
hdfs dfs -put local_dataset.csv /data/ingest/
hdfs dfs -ls /data/ingest/
hdfs dfs -cat /data/ingest/local_dataset.csv | head
```

Assignment 2: Data preprocessing & cleaning with Spark

```
from pyspark.sql import SparkSession
spark = SparkSession.builder.appName("clean").getOrCreate()
df = spark.read.csv("hdfs:///data/ingest/local_dataset.csv", header=True,
inferSchema=True) # handle missing values
train, test = df.randomSplit([0.8, 0.2], seed=42)
print("Train rows:", train.count(), "Test rows:", test.count())
```

Assignment 3: Word count with MapReduce (PySpark RDD)

```
text = spark.sparkContext.textFile("hdfs:///data/ingest/book.txt")
counts = (text.flatMap(lambda l: l.split())
           .map(lambda w: (w.lower(), 1))
           .reduceByKey(lambda a, b: a + b))
for word, c in counts.takeOrdered(10, key=lambda x: -x[1]):
    print(word, c)
```

Assignment 4: Data analysis with Spark MLlib

```
from pyspark.ml.classification import LogisticRegression
from pyspark.ml.feature import VectorAssembler
va = VectorAssembler(inputCols=["f1", "f2", "f3"], outputCol="features")
data = va.transform(df).select("features", "label")
model = LogisticRegression().fit(data)
print("Coefficients:", model.coefficients)
```

Assignment 5: Spark SQL query

```
df.createOrReplaceTempView("sales")
spark.sql("SELECT region, SUM(amount) AS total FROM sales GROUP BY region ORDER BY total DESC").show()
```

Assignment 6: Case study

Pick a domain (e.g. Netflix / Uber / Banking from the syllabus list), load its dataset into Spark, run exploratory analysis and one ML model, then present findings. Refer to 'Big Data Case Study' by Bernard Marr.

